ENERGY 01

Vitamin B12

Cobalt-containing vitamin (cobalamin); cyanocobalamin is the most widely used form



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
Vitamin	B12 deficiency; energy	FDA approved; 2 small trials for energy	FDA approved (deficiency) · energy off-label

STUDIED FOR

B12 deficiency	Chronic fatigue	Vocal fatigue
HUMAN	HUMAN	HUMAN

Energy data are limited. Two small placebo-controlled trials of B12 injections found no difference from placebo. No trial for weight loss was found.

HOW IT WORKS IN THE BODY

Vitamin B12 ▶ Carrier proteins ▶ Cells and liver ▶ Blood & myelin

- **Label:** essential to growth, cell reproduction, blood formation and myelin synthesis
- **Moved** to tissues by transcobalamin I and II; stored mainly in the liver
- **Hydroxocobalamin** works equally well, the label says

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- FDA approved injection for B12 deficiency
- 2 small placebo trials for energy or fatigue
- No trial of B12 injections for weight loss

NOT YET ESTABLISHED

- Energy effect without deficiency
- Weight-loss effect
- Compounded blends (not approved)

QUICK PRACTICAL INFO

HALF-LIFE	ROUTES STUDIED
Not established (not stated on the label)	Injection into muscle or under the skin (label)
PHARMACOKINETICS	**SOLD AS**
Peak within 1 hour (muscle); stored in liver; excreted in urine	Prescription injection (generic cyanocobalamin)
MONITORED IN TRIALS	**SPORT (WADA)**
Potassium, hematocrit, reticulocytes, B12, folate, iron	Not on the 2026 list

REGULATORY

FDA	Approved injection for B12 deficiency
FDA	Compounded methylcobalamin, MIC blends: not approved
WADA	Not listed; IV over 100 mL per 12 h banned (M2.2)

REFERENCES

1. FDA (DailyMed), Cyanocobalamin Injection USP label, Oct 2023
2. JAMA Otolaryngol Head Neck Surg, B12 injection vs placebo in singers, Jan 2021
3. WADA, 2026 Prohibited List

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